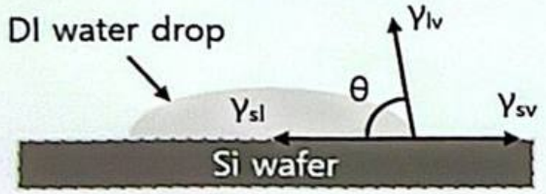
3 Effect of O_2 Plasma Activation on Surface Wettability

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Contact Angle Principle



Schematic of Water Contact Angle on a Si Wafer Surface

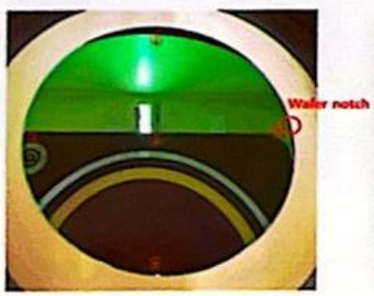
$$\gamma_{sv} = \gamma_{sl} + \gamma_{lv}\cos\theta$$

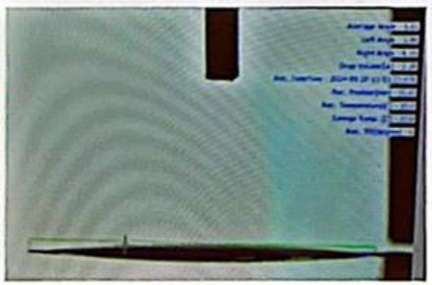
Water Contact Angle Measurement

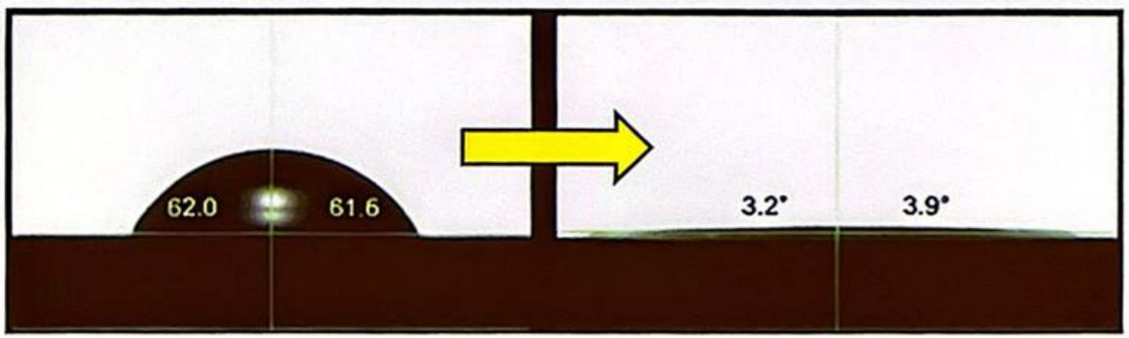
Probe liquid: DI Water
Comparison: Before/After Plasma Activation

RF		MF		O_2 Flow Rate (sccm)	Pressure (mTorr)
Power (W)	Freq. (MHz)	Power (W)	Freq. (kHz)		
168	13.56	100	25	100	250

Contact Angle Result







Plasma activation significantly reduced the water contact angle, indicating enhanced surface wettability and hydrophilicity.

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3 Dielectric-Dielectric Sample Preparation

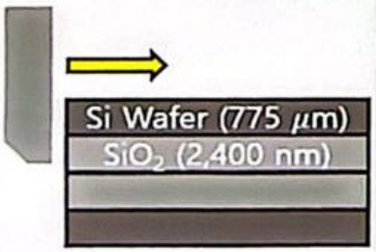
Justem

Dielectric-Dielectric Bonding Sample

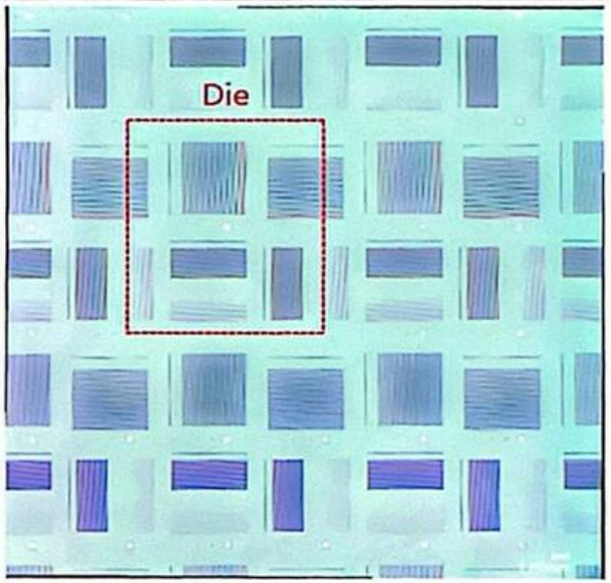
- ✓ Si Wafer: Samsung Trench Type 12 inch
- ✓ Die: Thinkness (777.4 μm), Size (14.6 x 12.8 mm)
- ✓ SiO_2 Deposition (2,400 nm)
- ✓ Plasma Treatment
  O_2 (RF 168 W, MF 100 W, 60 sec)
- ✓ Bonding
  R.T. Bonding
- ✓ Annealing
  Condition: 350 ℃, 2 h, 5,000 mbar

Interfacial Adhesion Test



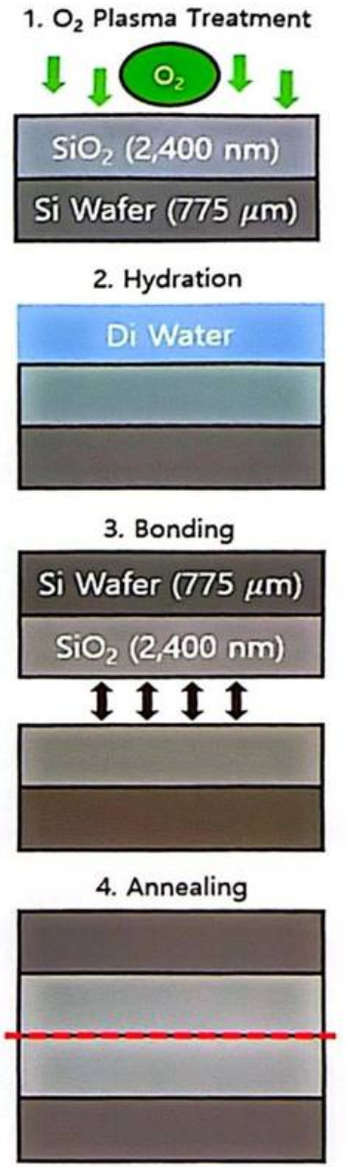


Dielectric Test Wafer Design



Trench Etch

- Wafer Diameter: 300 mm
- Trench Width: 60-200 nm in 5 nm Increments
- Bar (Active) Width: 500 and 1,000 nm
- Hole Width: 90-200 nm in 5 nm Increments
- Repeating Pattern Unit Size: 100 μm x 5,000 μm; Additional Patterns Rotated by 90°
- Si/SiO_2 Wafer; Thinkness: 2,400 nm
- Photoresist Ashing and Stripping Completed



※ Samsung test wafer purchased through the Korea Semiconductor Industry Association (KSIA).

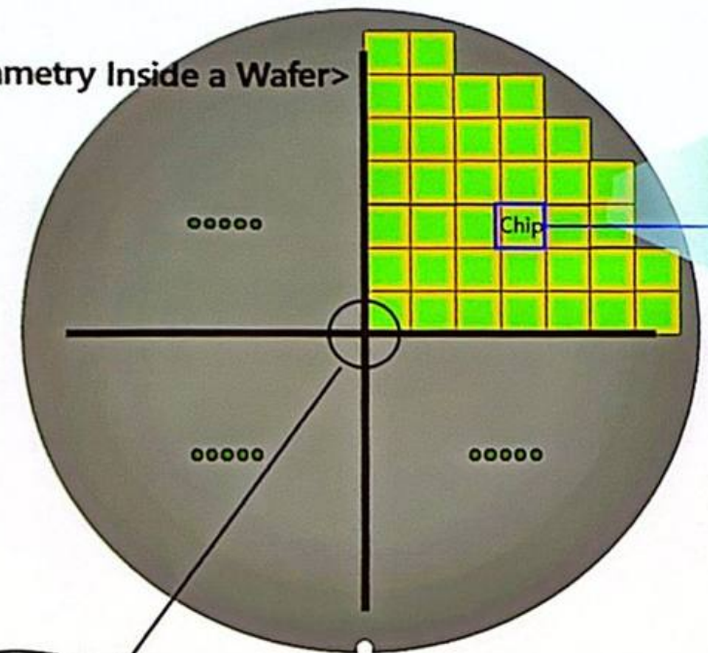
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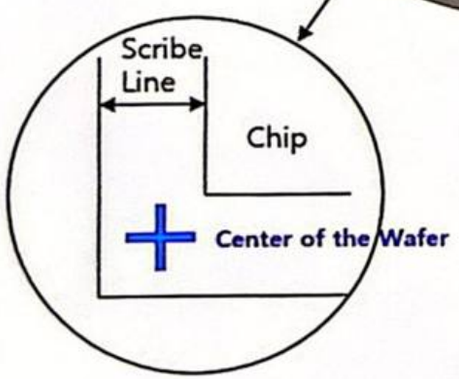
4 Design of a Hybrid Bonding Test Wafer

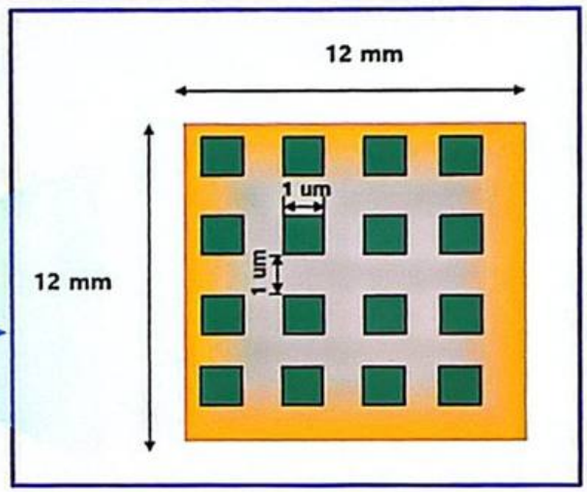
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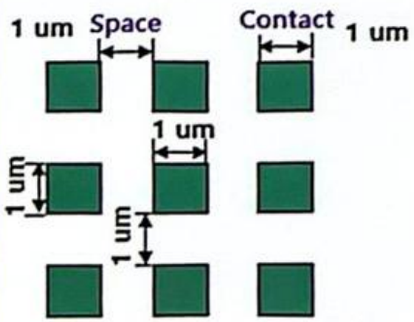
Cu/SiO_2 Patterned Wafer Structure and Fabrication

<4-Fold Symmetry Inside a Wafer>

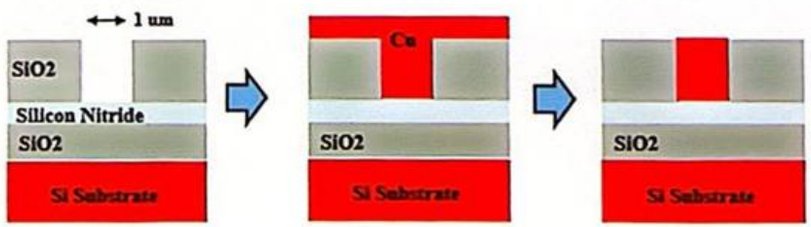








- Scribe Line Width of 100 μm
- Repetition Unit of 12 mm (∵Scribe Line Shared)
- Bonding Evaluation can be Proceeded with a Single Reticle and Identical Wafers.



- Thickness of SiO_2 and Silicon Nitride, Seed Metal and Cu

  ➔ Cu Damascene Standard Process Mask was Provided by Justem.

※ Cu/SiO_2 test wafer procured through the NNFC project (Project No. K26-002-S-PT)

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4 Hybrid Bonding Sample Preparation

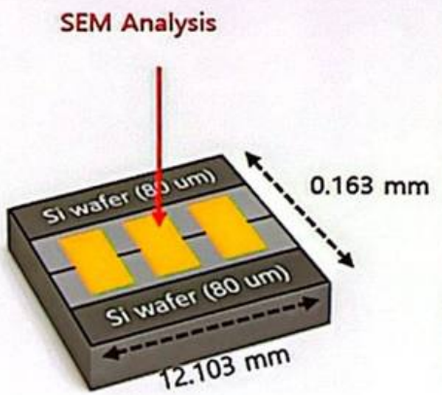
Justem

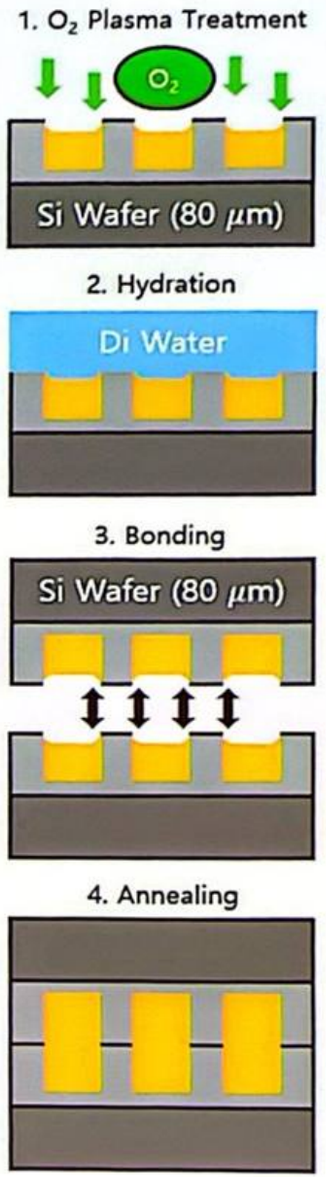
Cu/SiO_2 Hybrid Bonding Conditions

- Cu/SiO_2 Patterned Wafer
  SiO_2 Thickness: 1,500 nm
  Cu Pad Size: 1 μm x 1 μm
- O_2 Plasma: RF 168 W, MF 100 W, 60 sec
- Pre-bonding: Room Temperature
- Post-bond Annealing: 250 ℃, 300 ℃, 350 ℃ for 2 h

Cross-sectional SEM Analysis





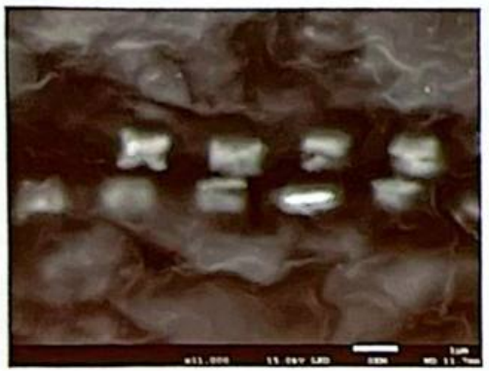


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4 Effect of Annealing Temperature on Cu-Cu Bonding Interface

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Cross-Sectional SEM Results



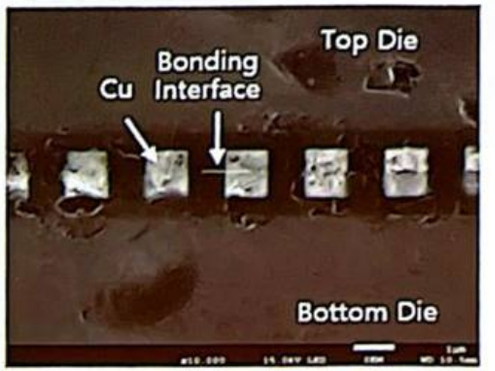
Annealing: 250 ℃, 2 h

Incomplete Cu–Cu Contact



Annealing: 300 ℃, 2 h

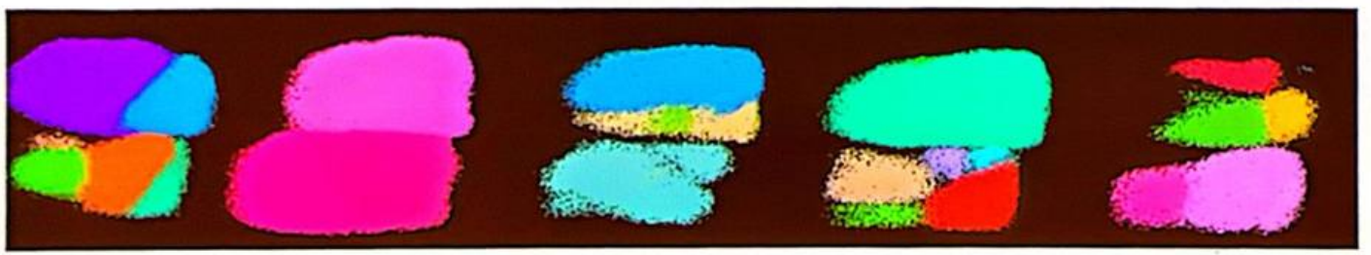
Continuous Cu–Cu Interfacial Contact

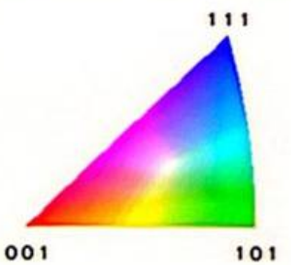


Annealing: 350 ℃, 2 h

Interface Deformation

Cross-Sectional EBSD Results





Annealing: 350 ℃, 2 h

Interface Deformation

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T29 D2D 하이브리드 본딩의 표면과 열처리

Byungchan Jung · JUSTEM

